Raynham Estate OS — North Star PRD & Design System

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Purpose of this document: This is the single source of truth for designing and building Raynham Estate OS. It is written to be read by both humans and AI development platforms (IDE copilots, AI app builders). Every screen, component, and decision must trace back to this document.

0. Bootstrap Prompt (Paste This Into Your AI Builder First)

```
You are building "Raynham Estate OS" — a premium, streamlined estate operations platform for a rural English country estate. It is a Progressive Web App (PWA), mobile-first for field workers and desktop-first for managers. It does exactly three jobs: (1) keep people safe via phone-based lone-worker check-ins and an emergency flow, (2) let anyone report anything by scanning a QR tag or dropping a map pin with a photo, and (3) give management one live spatial map of the whole estate showing staff, open reports, and active notices.

Brand identity: heritage-meets-modern. Warm off-white surfaces, slate-grey typography, and a muted brass gold (#C9A227) used sparingly as the accent. Light mode and dark mode are both required, driven by design tokens. The interface must be calm, uncluttered, and usable by older field staff: large touch targets, plain language, minimal navigation (Live Map, Reports, Safety, Notices, Settings), and three workspace tabs (Farm & Land, Property & Grounds, Hospitality).

Every location displays a What3Words address alongside GPS. The product is explicitly NOT an enterprise dashboard: no dense data grids, no complex charts, no compliance theatre. Build the design token system first, then the component library, then screens. All tokens, components, and screen specs are defined in the accompanying PRD, which overrides any assumption you might otherwise make.
```

1. Product Overview

1. What it is: Raynham Estate OS is a white-labelled, stripped-down edition of the ADOS Genesis spatial platform, tailored to a single rural estate with multiple business ventures (arable farm, livestock, forestry, property maintenance, glamping, pop-up restaurant, hall events, commercial and residential lettings).
2. What it does: It connects the physical estate to one digital map using weatherproof QR anchors on assets, buildings, vehicles, and infrastructure. Staff, tenants, and contractors scan a tag (no app download, no login for reporters) to get information, report an issue with a photo, or request emergency help.
3. What it replaces: A physical lone-worker fob system (fobs get left in vehicles; phones don't), and a manual reporting chain where management acts as a middleman between the person who finds a problem and the person who fixes it.
4. What it is not: It is not a back-office system. It does not touch accounting (Xero), farm management software, or stock control. It is a field tool for operators, not an oversight system for data.

2. The Three Jobs (Every Feature Must Serve One)

1. Keep people safe — every worker's phone is the safety beacon. Check-ins, lone-worker timers, and a red emergency flow. Coverage extends to all staff and contractors, not just high-risk roles.
2. Report anything — scan a QR tag or drop a pin, take a photo, write or dictate a note, done. The system routes it to the right team or contractor automatically.
3. See the estate — one spatial map with every open job, active notice, and checked-in worker. The estate owner sees everything; each manager sees their workspace.

Rule: if a proposed feature does not clearly serve one of these three jobs, it is out of scope.

3. Design Principles

1. Calm by default. The interface only raises its voice when something is wrong. One red pulsing marker commands more attention than a wall of alerts.
2. Speak the estate's language. What3Words addresses appear on every pin, anchor, report, and alert. The team already uses them; the system adopts their habit rather

than teaching a new one.

3. Field-first ergonomics. Designed for gloved hands, bright sunlight, patchy GSM signal, and users who will not read manuals. Minimum 44px touch targets, high contrast, offline queueing.
4. Three taps maximum. Any field action (report, check-in, emergency) must complete in three taps or fewer from a QR scan.
5. Heritage, not corporate. The aesthetic belongs to a country estate: warm whites, slate, brass gold — never startup-blue, never military theatre.
6. Progressive disclosure. Field staff see three buttons. Managers see a dashboard. The owner sees the estate. Complexity is revealed by role, never all at once.

4. Scope

In scope:

1. QR anchor registry and bi-directional tagging (Active Notices on scanned assets)
2. Three-choice mobile scan interface: Information (green) / Report (amber) / Get Help Now (red)
3. Photo-based reporting with auto-captured GPS + What3Words address
4. Lightweight AI routing: auto-categorise each report and assign to the correct workspace/team
5. Contractor assignment loop via email/SMS with resolution tracking
6. Lone-worker check-ins with timed sessions and escalation
7. Live estate map (satellite) with pins for staff, reports, alerts, and anchors
8. Notices board with templates and mandatory auto-expiry
9. Executive overview ("Tom's View"): whole-estate map + quiet KPI tiles
10. Light mode and dark mode, token-driven
11. PWA: installable, offline queue, GSM-first

Explicitly out of scope (do not build):

1. Any revenue/engagement layer (no offers, no sponsored content)
2. Compliance-framework features (no Martyn's Law positioning; the timestamped record exists quietly as a by-product)
3. AI voice agents, autonomous task swarms, auto-Kanban generation
4. Integrations with Xero, farm management software, ServiceNow, Salesforce, or any back-office system
5. Per-user licence gates — the model is a flat estate licence with unlimited users; the

app must not restrict onboarding

6. Guest-facing hospitality features beyond a simple info page (Phase 3 decision, owner's choice)
7. Dense analytics/charting — three KPI tiles are the ceiling

5. Users & Roles

1. Owner (Tom): full visibility across all workspaces; executive view by default; manages data export and billing contacts.
2. Workspace Manager (3 seats: Farm & Land, Property & Grounds, Hospitality): sees their workspace's reports, notices, and team; assigns jobs to staff or contractors; publishes notices.
3. Field Staff: scan anchors, report issues, check in as lone workers, receive Active Notices. No login required to scan and report; a lightweight profile (name + phone) enables check-ins.
4. Contractors (external): receive assigned jobs by email/SMS with photo, What3Words location, and a one-tap "mark resolved" link. No account.
5. Tenants: report property issues via the anchor QR in their home. No account.

6. Information Architecture

1. Three workspaces (top-level filter, styled as tabs): Farm & Land · Property & Grounds · Hospitality. Workspaces share one map, one reports engine, one notices engine — they are lenses, not silos.
2. Five primary destinations (left nav rail on desktop, bottom bar on mobile): Live Map · Reports · Safety · Notices · Settings.
3. Hierarchy for the spatial registry: Estate → Workspace → Zone → Raynham Estate → Farm & Land → North Field → Anchor "Gat
4. Default landing: Live Map, filtered to the signed-in user's workspace; the owner lands on the whole-estate view.

7. Core Features

7.1 QR Anchors & Bi-Directional Tagging

1. Each anchor is a weatherproof physical QR tag registered to a zone and optionally an asset (tractor, yurt, pump, property).

2. Scanning an anchor opens the three-choice interface with the anchor's identity and What3Words address pre-loaded.
3. Bi-directional Active Notices: when an anchor has an open report or scheduled check, anyone who scans it sees the notice first — e.g. "Martin is due to check this machinery at 4 p.m. Please do not use until verified."
4. Anchors work offline at the point of scan: the QR encodes a URL; the page caches the anchor identity; actions queue when signal drops and sync on reconnection.

7.2 Reporting (Amber Flow)

1. From scan or map pin: capture photo (required option), pick a category (chips, max 8, workspace-specific), add an optional note (text or voice-to-text), submit.
2. System auto-attaches: GPS, What3Words address, timestamp, anchor identity, reporter (if known).
3. AI routing (light-touch): on submit, classify the report and route it to the owning workspace and team; flag duplicates and repeat locations. No visible "AI" branding — it simply arrives in the right inbox.
4. Report lifecycle states: New ’ Assigned ’ In Progress ’ Res optional). Each transition is timestamped.
5. Managers assign to internal staff or external contractors (email/SMS loop). Resolution requires a confirmation tap; photo optional.

7.3 Safety (Red Flow + Lone Worker)

1. Get Help Now: one oversized red button from any scanned anchor or the app's persistent safety tab. Confirm with a single tap (prevents false alarms, adds no friction). Dispatches an alert to the on-call manager with GPS + What3Words + reporter identity.
2. Lone-worker sessions: worker starts a session (task + expected duration, e.g. "Chainsaw work, 90 min"). If the session expires without check-out, the system alerts the on-call manager with the last known location.
3. Universal coverage: no hardware. Any staff member or contractor with a phone can start a session — this is the direct replacement for the fob system.

7.4 Notices (Green Flow + Publishing)

1. Anchors display informational content: site rules, opening info, wayfinding,

temporary guidance.

2. Managers publish from templates (Facilities · Delay · Closure · Safety · Custom) with a mandatory expiry date/time. Expired notices disappear automatically — nothing goes stale.

7.5 Live Map

1. Satellite base map of the estate with four pin types: staff check-in (green), open report (amber), active alert (red, pulsing), registered anchor (gold dot).
2. Every pin's detail card shows its What3Words address, with a copy button and tap-to-navigate.
3. Workspace tabs filter all pins and panels. The owner's "All" view overlays everything.
4. Pin clustering at low zoom; individual pins resolve as you zoom in.

7.6 Executive View (Tom's View)

1. Whole-estate map, all workspaces, all live activity.
2. Exactly three KPI tiles: Staff checked in · Open reports · Active notices.
3. A weekly plain-language summary generated from report data (what happened, what's recurring, what's overdue). One screen, no charts beyond the three tiles.

8. Key User Flows

1. Report an issue (field): Scan anchor !’ amber "Report" !’ ph
   submit. Three taps after scan. Confirmation shows the What3Words reference.
2. Report without an anchor: Open app !’ Live Map !’ long-pres
   flow. System snaps the pin to the nearest zone and generates the What3Words address.
3. Contractor resolution: Receives SMS/email !’ opens link !’ s
   three-word address !’ taps "Mark resolved" !’ manager notif
4. Lone worker: Safety tab !’ "Start session" !’ task + duration
   !’ "Check out." Expiry !’ manager alert.
5. Emergency: Any anchor scan !’ red "Get Help Now" !’ single
   dispatched. Also available from the app's persistent red tab without scanning.
6. Machinery notice: Worker reports broken tractor !’ manager
   that anchor !’ next person to scan the tractor sees the not

9. Design Language & Theme Direction

1. Light mode (default, "Estate"): warm off-white paper surfaces, slate-grey text, brass-gold accents, thin warm-grey borders, soft low-opacity shadows. Satellite imagery is the hero.
2. Dark mode ("Night Shift"): deep warm charcoal (not pure black), the same gold at a brightened value for contrast, glowing accents reserved for the alert pulse. For early mornings, vehicle cabs, and night checks.
3. Both modes are one system, two skins. All colour decisions go through semantic tokens; components never hard-code hex values.
4. Gold is scarce. It marks the brand, the active workspace, and anchor pins. Overuse is a design defect.
5. Status colours are universal across both modes and match the physical world's conventions: green = safe/resolved, amber = attention/in progress, red = emergency only.

10. Design Tokens

10.1 Colour Tokens — Light Mode (CSS Custom Properties)

```
:root {
  /* Surfaces */
  --surface-base:    #FAFAF7;  /* warm paper off-white */
  --surface-raised:  #FFFFFF;  /* cards, panels */
  --surface-sunken:  #F3F2EC;  /* wells, inputs */
  --surface-overlay: rgba(250, 250, 247, 0.92); /* floating panels over map */

  /* Borders */
  --border-subtle:   #E6E4DB;
  --border-strong:   #D4D1C4;

  /* Text */
  --text-primary:    #2E3438;  /* deep slate */
  --text-secondary:  #5B6770;  /* brand slate (wordmark grey) */
  --text-muted:      #8A9299;
  --text-inverse:    #FFFFFF;

  /* Brand */
```

```
  --brand-gold:      #C9A227;  /* primary accent — use sparingly */
  --brand-gold-hover:#B08D1E;
  --brand-gold-soft: #F5EBCE;  /* gold tint fill */
  --brand-slate:     #5B6770;

  /* Status */
  --status-success:  #2F7D4F;  /* safe / resolved / checked in */
  --status-success-soft: #E3F0E7;
  --status-warning:  #C9821E;  /* attention / in progress */
  --status-warning-soft: #F9EEDC;
  --status-danger:   #C43D2E;  /* EMERGENCY ONLY */
  --status-danger-soft:  #F8E3DF;
  --status-neutral:  #6B7280;  /* assigned / informational */
  --status-neutral-soft: #ECEEF0;

  /* Map pins */
  --pin-staff:   #2F7D4F;
  --pin-report:  #C9821E;
  --pin-alert:   #C43D2E;
  --pin-anchor:  #C9A227;

  /* Focus & interaction */
  --focus-ring:  rgba(201, 162, 39, 0.45);
  --scrim:       rgba(18, 20, 22, 0.48);
}
```

10.2 Colour Tokens — Dark Mode

```
[data-theme="dark"] {
  --surface-base:    #16181B;  /* warm charcoal, not pure black */
  --surface-raised:  #1E2126;
  --surface-sunken:  #101215;
  --surface-overlay: rgba(30, 33, 38, 0.92);

  --border-subtle:   #2B2F35;
  --border-strong:   #3A3F46;

  --text-primary:    #F2F1EC;
  --text-secondary:  #A9B2B8;
  --text-muted:      #7C848B;
```

```
  --text-inverse:    #16181B;

  --brand-gold:      #D9B545;  /* brightened for dark-surface contrast */
  --brand-gold-hover:#E4C55F;
  --brand-gold-soft: #3A3319;
  --brand-slate:     #A9B2B8;

  --status-success:  #4CAF6E;
  --status-success-soft: #1E3327;
  --status-warning:  #E0A03C;
  --status-warning-soft: #3B2F16;
  --status-danger:   #E35D4D;
  --status-danger-soft:  #3E211D;
  --status-neutral:  #9AA3AB;
  --status-neutral-soft: #2B2F35;

  --pin-staff:   #4CAF6E;
  --pin-report:  #E0A03C;
  --pin-alert:   #E35D4D;
  --pin-anchor:  #D9B545;

  --focus-ring:  rgba(217, 181, 69, 0.5);
  --scrim:       rgba(0, 0, 0, 0.6);
}
```

10.3 Typography, Spacing, Radius, Shadow, Motion (JSON Token File)

```
{
  "font": {
    "family": {
      "ui":    { "value": "Inter, system-ui, sans-serif" },
      "brand": { "value": "Inter, system-ui, sans-serif", "note": "Rendered uppercase wit
tracking for wordmarks/eyebrows" },
      "mono":  { "value": "'IBM Plex Mono', ui-monospace, monospace", "note":
"What3Words addresses and reference codes" }
    },
    "size": {
      "display": { "value": "2rem",    "lineHeight": "2.5rem" },
      "h1":      { "value": "1.5rem",  "lineHeight": "2rem" },
      "h2":      { "value": "1.25rem", "lineHeight": "1.75rem" },
```

```
      "h3":      { "value": "1rem",    "lineHeight": "1.5rem" },
      "body":    { "value": "0.9375rem", "lineHeight": "1.375rem" },
      "small":   { "value": "0.8125rem", "lineHeight": "1.125rem" },
      "micro":   { "value": "0.6875rem", "lineHeight": "1rem", "letterSpacing": "0.08em",
"textTransform": "uppercase" }
    },
    "weight": {
      "regular": { "value": "400" },
      "medium":  { "value": "500" },
      "semibold":{ "value": "600" }
    }
  },
  "spacing": {
    "1":  { "value": "0.25rem" },
    "2":  { "value": "0.5rem" },
    "3":  { "value": "0.75rem" },
    "4":  { "value": "1rem" },
    "5":  { "value": "1.25rem" },
    "6":  { "value": "1.5rem" },
    "8":  { "value": "2rem" },
    "10": { "value": "2.5rem" },
    "12": { "value": "3rem" },
    "16": { "value": "4rem" }
  },
  "radius": {
    "sm":   { "value": "6px" },
    "md":   { "value": "10px" },
    "lg":   { "value": "16px" },
    "pill": { "value": "999px" }
  },
  "shadow": {
    "card-light":  { "value": "0 1px 2px rgba(46,52,56,0.06), 0 4px 12px
rgba(46,52,56,0.06)" },
    "panel-light": { "value": "0 8px 24px rgba(46,52,56,0.10)" },
    "card-dark":   { "value": "0 1px 2px rgba(0,0,0,0.4), 0 4px 12px rgba(0,0,0,0.3)" },
    "alert-glow":  { "value": "0 0 0 4px rgba(196,61,46,0.25), 0 0 24px
rgba(196,61,46,0.45)" }
  },
  "motion": {
    "fast":   { "value": "120ms" },
    "base":   { "value": "200ms" },
    "slow":   { "value": "320ms" },
```

```
    "easing": { "value": "cubic-bezier(0.2, 0, 0, 1)" },
    "alertPulse": { "value": "1600ms infinite", "note": "Reserved exclusively for the red
emergency state" }
  },
  "layout": {
    "navRailWidth":   { "value": "72px" },
    "sidePanelWidth": { "value": "360px" },
    "headerHeight":   { "value": "64px" },
    "touchTargetMin": { "value": "44px" },
    "maxContentWidth":{ "value": "1440px" }
  },
  "breakpoints": {
    "mobile":  { "value": "0–767px" },
    "tablet":  { "value": "768–1023px" },
    "desktop": { "value": "1024px+" }
  }
}
```


10.4 Token Rules

1. Components consume semantic tokens only (--surface-raised, --status-warning), never raw palette values.
2. --status-danger and --pin-alert may only be used for the emergency flow and active alerts. Using red anywhere else is a defect.
3. Dark mode is applied by toggling data-theme="dark" on the root element; the user's choice persists per device.
4. The --motion-alertPulse animation is the only infinite animation in the system.

11. Design System Components

11.1 Buttons

1. Primary: gold fill (--brand-gold), dark-slate text, 10px radius, height 44px (mobile) / 36px (desktop). Used for the single main action per screen.
2. Secondary: transparent fill, 1px --border-strong border, slate text.
3. Danger: --status-danger fill, white text. Reserved for "Get Help Now" and "End session". Full-width on mobile, height 56px.
4. Ghost: no fill or border, slate text, for tertiary actions.

5. States: hover (gold ’ --brand-gold-hover), pressed (scale 0
   opacity), loading (spinner replaces label).

11.2 Status Chips

1. Pill shape, 12px text, soft background + solid text: New (neutral), Assigned (neutral), In Progress (warning), Resolved (success), Alert (danger, solid fill).
2. Chips are read-only indicators; state changes happen through actions, never by tapping the chip.

11.3 Report Card

1. Anatomy: 64px photo thumbnail · plain-language title (e.g. "Broken fence — North Field") · What3Words address in mono font · team tag · status chip · relative timestamp ("2h ago").
2. Tapping opens the Report Detail: photo, notes, map snippet, timeline of state changes, assign/resolve actions.

11.4 KPI Tile

1. White/raised card, thin border, small gold-or-status icon, large display numeral, micro uppercase label. Exactly three exist: Staff checked in · Open reports · Active notices.

11.5 Map Pins

1. Anchor: 10px gold dot with white ring — quiet, always present.
2. Staff check-in: 14px green pin with initials.
3. Report: 16px amber pin with category glyph.
4. Alert: 18px red pin with the --motion-alertPulse glow. The only animated element on the map.
5. Tapping any pin opens its detail card in a bottom sheet (mobile) or side panel (desktop).

11.6 What3Words Chip

1. Mono font, ///three.word.address format, copy icon, tap-to-navigate (opens device maps). Appears on every pin card, report, alert, and anchor page.

11.7 Active Notice Banner

1. Rendered at the top of any scanned anchor page when a notice is live: amber left border, plain-language message, expiry time. Dismissible only by the notice's owner or a manager.

11.8 Workspace Tabs

1. Three tabs; active tab underlined in gold with slate text; inactive tabs in muted grey. Persist selection across sessions per user.

11.9 Navigation Rail (Desktop) / Bottom Bar (Mobile)

1. Five destinations with outlined icons + micro labels: Live Map · Reports · Safety · Notices · Settings.
2. Safety always carries a subtle red icon tint; all others are slate. Active destination gets a gold indicator bar.

11.10 Three-Choice Scan Panel (Mobile)

1. The post-scan landing: three full-width stacked cards, 72px tall each — Information & Updates (green accent), Report a Concern (amber accent), Get Help Now (red accent) — followed by the anchor identity and What3Words chip.

11.11 Session Card (Lone Worker)

1. Shows task, elapsed/remaining time, and one "Check out" danger button. Persistent on the worker's device during the session; sends a local notification at 10 minutes to expiry.

11.12 Supporting Primitives

1. Bottom sheet (mobile), modal (desktop), toast (4s, top on mobile / bottom-right desktop), empty states (one line of guidance + one action), skeleton loaders for map panels, offline banner ("No signal — reports will send automatically when reconnected").

12. Screen Specifications

1. Live Map (default): header (wordmark, date, GSM indicator, theme toggle, "All Clear" status pill) · nav rail with workspace tabs · satellite map with pins · right "Open Reports" panel (desktop) · three KPI tiles bottom-left.
2. Reports: filterable inbox (status, workspace, team) sorted by newest; bulk assign; report detail view with timeline.
3. Safety: live lone-worker sessions with countdowns; alert history; on-call contact management.
4. Notices: active notices list with expiry countdowns; "New notice" composer (template → anchor/zone → message → expiry → publish).
5. Settings: workspace management, anchor registry (add/print/reassign QR tags), team management, data export (CSV/PDF), theme preference.
6. Mobile scan landing: Three-Choice Scan Panel + Active Notice (if any) + anchor identity.
7. Mobile report flow: camera-first capture → category chips → confirmation with What3Words reference.
8. Mobile safety: persistent red tab; "Get Help Now" confirm sheet; "Start lone-worker session" composer.
9. Executive view: whole-estate map, all-workspace overlay, three KPI tiles, weekly plain-language summary card.
10. Auth: magic-link email for managers/owner; field staff identified by name + phone on first use; reporters/tenants/contractors need no account.

13. Data Model (Core Entities)

1. Estate → Workspace (3) → Zone → Anchor (QR payload URL, asset link).
2. Asset (optional): machinery, vehicle, property, yurt, infrastructure.
3. Report: photos, category, note, reporter, location (lat/lng + W3W), anchor ref, status, assignee (internal or contractor contact), timestamps.
4. Notice: template type, body, scope (anchor/zone/workspace), expiresAt (mandatory), author.
5. Session (lone worker): user, task, startedAt, expiresAt, checkedOutAt, lastLocation.
6. Alert: type (help/session-expiry), user, location + W3W, triggeredAt, acknowledgedBy, resolvedAt.
7. User: role (owner/manager/staff), workspace memberships, phone.
8. Audit trail: every status transition on every entity is timestamped and stored — this

is the quiet evidence record.

14. Technical Architecture

1. PWA, mobile-first: installable to home screen, service worker caching, offline report queue with background sync. GSM-first — must function on 3G; degrade gracefully, never block.
2. Suggested stack: Next.js (React) + Tailwind CSS (tokens mapped to CSS variables) + MapLibre GL (or Mapbox) satellite basemap + Supabase (Postgres, Auth, Storage for photos, Realtime for live pins) + Vercel hosting.
3. What3Words API: convert coordinates!'address on pin creat registration; cache results server-side so addresses display offline.
4. QR anchors: generated per anchor, encoding the canonical URL /a/{anchorId}; printable sheet export for physical tag production.
5. AI routing: single classification step on report submission workspace/team mapping with confidence; fall back to "unassigned" queue). One model call, no agent framework.
6. Notifications: SMS + email via a transactional provider for contractor loops and safety alerts; push notifications for managers.
7. Data ownership (contractual, must be honoured in build): the estate owns 100% of its data; full export (CSV/PDF) available in Settings at any time; complete deletion on exit; no third-party training on estate data; UK/EU hosting.

15. Accessibility & Field-Use Requirements

1. WCAG 2.2 AA contrast in both themes; never communicate state by colour alone (icons + labels accompany status colours).
2. Minimum 44px touch targets; primary field flows operable one-handed and with work gloves.
3. All field flows functional at 3G speeds; photo uploads compress and queue.
4. Plain language throughout — no jargon, no feature names on user-facing surfaces ("Report a Concern", never "ARIA Triage").
5. Light mode optimised for outdoor sunlight legibility; dark mode for low-light vehicle use.

16. Phased Build Roadmap

1. Phase 1 — Foundation (Farm & Land + Safety): tokens, component library, map, anchors, reporting, lone-worker safety, notices. Directly replaces the fob system.
2. Phase 2 — Property & Grounds: tenant reporting anchors on residential properties, contractor assignment loops, maintenance team workflows.
3. Phase 3 — Hospitality (pre–school holidays): glamping/restaurant/hall staff workflows; optional guest-facing info pages only if the owner requests them.

17. Success Metrics

1. Weekly active field users vs. total staff (target: 80% by end of Phase 1).
2. Median time from report ’ assignment (target: under 1 work
3. Lone-worker session coverage: % of high-risk tasks started with an active session.
4. Cost per covered worker vs. the legacy fob system (the headline commercial comparison).

18. Open Questions for the Client

1. Guest-facing hospitality layer in year one — yes or not yet?
2. Preferred SMS/email provider constraints, if any existing contracts exist.
3. Which single manager is "on call" for safety alerts per workspace, and the escalation order.

This document is the North Star. If reality and this document disagree, update the document — don't fork the product.